

Lecture Notes

Training Methods for LLMs: Pre-training and Backpropagation

What is an LLM?

An **LLM (Large Language Model)** is fundamentally a sophisticated mathematical function designed to predict the most statistically probable next word in any sequence of text. Its entire operation revolves around probability, assigning a likelihood score to every possible word following the input prompt.

The Core Processes: Pre-training and Backpropagation

- **Pre-training:** This is the initial, massive phase where the model learns general language patterns by consuming colossal amounts of raw data (e.g., the entire internet). The primary goal is to make the model an expert at **auto-completion**.
- **Backpropagation:** This is the mathematical algorithm—the core engine—that allows the model to learn from its mistakes and systematically adjust its internal knowledge over time, improving future predictions.

Why Are These Methods Necessary?

These complex training methods are required for two distinct purposes: building foundational knowledge (**Scale**) and ensuring usability (**Refinement**).

1. **Building Knowledge (Pre-training):** By forcing the model to predict the next word trillions of times, it absorbs vast amounts of structured data—grammar rules, historical facts, coding syntax, and cultural patterns—without needing explicit instruction for each fact.
2. **Ensuring Usability (Refinement):** Raw pre-trained models are often biased or unhelpful. Techniques like **Reinforcement Learning with Human Feedback (RLHF)** are used to *align* the model, teaching it not just what is statistically correct, but what is safe, helpful, and conversational according to human standards.

How Does the Training Work?

The process moves through three interconnected stages: encoding text, establishing architecture, and optimizing knowledge.

1. The Foundation: Encoding Text (Vectors)

Since computers only understand numbers, every word or unit of text (**token**) must first be converted into a long list of numbers called a **vector**. This vector

attempts to encode the *meaning* and *context* of that token within the language.

2. The Architecture: Transformers and Attention

The **Transformer** architecture revolutionized LLMs by enabling **parallel processing**. Instead of reading text sequentially (like older models), it can process an entire chunk of text simultaneously, making training vastly more efficient.

- **Attention Mechanism:** This is the key function within the Transformer. When processing a word, the mechanism allows that word's vector to dynamically look at *all* other words in the input sequence and assign different levels of importance (weights).
 - **Example:** In “*The river bank was muddy,*” Attention ensures the model knows that “river” must modify the meaning of “bank” from a financial institution to a geographical feature, resolving ambiguity based on context.

3. The Learning Engine: Backpropagation in Action

This is the iterative optimization loop where learning occurs across trillions of data points:

1. **Input:** An example text sequence is fed into the model (e.g., “*The capital of France is...*”).
2. **Prediction:** The model processes this input and generates a probability distribution for the next word (e.g., “Paris” at 80%, “Rome” at 10%).
3. **Ground Truth:** The actual correct answer is provided (“Paris”).
4. **Error Calculation: Backpropagation** compares the model's prediction to the ground truth, calculating a massive **error signal**.
5. **Parameter Adjustment:** The algorithm then systematically adjusts *every single parameter* (the internal weights or “dials”) in the entire network. This adjustment is mathematically designed to increase the probability of the correct answer (“Paris”) and decrease the probabilities of all incorrect guesses.

By repeating this error correction process billions of times, the model's parameters become finely tuned to accurately predict contextually appropriate next words.

Key Concepts Summary

Term	Definition	Role in LLMs
Parameters/Weights	The numerical values that constitute the model's learned knowledge.	They are adjusted by Backpropagation; changing them changes the model's behavior.

Term	Definition	Role in LLMs
Transformers	An advanced neural network architecture allowing parallel processing of text.	Enabled efficient, contextual understanding across entire sequences.
Attention Mechanism	The function that assigns varying importance (weights) to different words in a context.	Allows ambiguity resolution and deep contextual understanding.
Pre-training	Initial phase using raw data for auto-completion.	Builds general world knowledge and grammar patterns.
Backpropagation	The core algorithm used to calculate error signals and adjust parameters.	Drives the learning process by correcting predictions based on mistakes.

Common Pitfalls (What Not To Say)

- **Prediction vs. Understanding:** LLMs are complex pattern-matching systems that calculate probabilities; they do not possess human understanding or consciousness.
- **Deterministic Output:** Due to random sampling (controlled by parameters like ‘temperature’), the model is non-deterministic and can produce varied answers even with identical prompts.

Summary of Functionality

The entire process relies on **Backpropagation** adjusting **Parameters/Weights** within a **Transformer** architecture, which uses the **Attention Mechanism** to predict the next most probable word based on vast amounts of data learned during **Pre-training**. This raw knowledge is then refined through techniques like RLHF for safety and utility. # Microbiology: Bacteria vs. Viruses

Understanding Pathogens: Bacteria vs. Viruses

This topic clarifies the fundamental differences between two major categories of infectious agents—**bacteria** and **viruses**—which are often mistakenly grouped together. Understanding these distinctions is critical for proper diagnosis and treatment.

Key Differences in Structure and Life Cycle

Feature	Bacteria	Viruses
Cellular Nature	Are true, single-celled organisms (prokaryotes).	Are not cells; they are packets of genetic material (DNA or RNA).
Reproduction	Can reproduce independently using their own machinery.	Must hijack a living host cell's machinery to replicate.
Structure	Contain cytoplasm, ribosomes, and cell walls.	Consist only of nucleic acid (genome) encased in a protein coat (capsid).
Size	Relatively large (visible with standard microscopes).	Extremely small (require electron microscopes).
Treatment	Targeted by antibiotics which kill or inhibit their cell wall/metabolism.	Treated with antivirals , which interfere with the viral life cycle <i>inside</i> the host cell.

Detailed Breakdown of Each Agent

Bacteria Bacteria are complete, self-sufficient organisms.

- **Structure:** They possess all necessary components for metabolism and reproduction (e.g., ribosomes, DNA).
- **Reproduction:** They reproduce asexually through **binary fission**, splitting into two identical daughter cells.
- **Example:** *Streptococcus* (causes strep throat), *E. coli*.
- **Treatment Principle:** Antibiotics target unique bacterial features, such as the peptidoglycan layer in their cell walls or specific metabolic pathways.

Viruses Viruses are obligate intracellular parasites. They cannot perform life functions on their own.

- **Structure:** A virus particle (virion) is minimal: a nucleic acid genome (DNA or RNA) enclosed by a protein shell (**capsid**). Some viruses also have an outer lipid envelope derived from the host cell.
- **Reproduction Cycle:** Viruses follow a mandatory cycle involving attachment, penetration, replication of their genetic material, synthesis of new proteins, assembly, and release.
- **Example:** Influenza virus (the flu), SARS-CoV-2 (COVID-19).
- **Treatment Principle:** Antivirals work by blocking specific steps in the viral life cycle (e.g., preventing the virus from entering the cell or assembling its components).

Clinical Significance: Why the Difference Matters

The distinction is medically vital because **antibiotics are completely ineffective against viruses**.

- If a person has a common cold, which is typically viral, taking antibiotics will not cure it and can contribute to antibiotic resistance in the patient's microbiome.
- Conversely, infections caused by bacteria (like urinary tract infections) require antibiotics for effective treatment.

Summary of Key Concepts

- **Bacteria:** Self-sufficient prokaryotes; treated with **antibiotics**.
- **Viruses:** Non-cellular parasites; must hijack host cells to replicate; treated with **antivirals**.
- **Core Principle:** The ability to reproduce independently defines the difference between these two groups.